

Question

We denote the position of the amino acid residue at the *C*-terminus of a peptide chain cleaved by a protease as 0, and sequentially forward as -1, -2, -3, and backward as +1, +2, +3, and so on. For example, the substrate recognized by trypsin is a residue at position 0 that is lysine or arginine, and the residue at position +1 is not proline.

Subtilisin is a serine/threonine protease. Subtilisin has the ability to recognize longer substrate series and has certain selectivity for different residues at positions -3 to 0. We denote the substrate pocket recognizing the 0 residue as S1, the pocket for the -1 residue as S2, -2 as S3, and -3 as S4.

In wild-type subtilisin, the S4 pocket residue is Tyr104, the S2 pocket is Asn62, and the S1 pocket is Gly166. Scientists tested the selectivity of wild-type subtilisin for different substrates using a C-terminally substituted tetrapeptide substrate XXXX-p-nitroanilide, and found that the fastest reacting sequence was Ala-Ala-Pro-Phe-p-nitroanilide.

Now we consider a batch of subtilisin mutants:

- A. G166D, N62D
- B. G166L, Y104D
- C. N62V, Y104K

And some substrates:

1. AAPF-p-nitroanilide
2. AAPA-p-nitroanilide
3. AAPQ-p-nitroanilide
4. AAPY-p-nitroanilide
5. DAKF-p-nitroanilide
6. RAKR-p-nitroanilide

7. RGKE-p-nitroanilide
8. DAPR-p-nitroanilide
9. DAVF-p-nitroanilide
10. FGKR-p-nitroanilide
11. LGFR-p-nitroanilide
12. AAKR-p-nitroanilide
13. RGAA-p-nitroanilide
14. DAHF-p-nitroanilide
15. RGAG-p-nitroanilide

Then, matching these mutants with the substrates with which they react fastest, which of the following options has the numbers from front to back representing the substrate numbers corresponding to subtilisin A, B, and C, respectively, with completely correct results?

A. 6, 13, 5

B. 6, 13, 9

C. 6, 13, 14

D. 6, 15, 5

E. 6, 15, 14

F. 10, 13, 5

G. 10, 13, 9

H. 10, 13, 14

I. 10, 15, 5

J. 10, 15, 14

K. 12, 13, 14

L. 12, 13, 5

M. 12, 15, 14

N. 12, 15, 9

O. All of the above options are incorrect.

Answer

Correct Answer: **0**

Detailed Explanation

The problem gives 14 different substrates. First, we roughly classify these substrates according to the residue type at different positions, as shown in the table below. Note that the order of pocket numbers and amino acid residue numbers is reversed. S4 corresponds to the first amino acid in the substrate.

Position	Hydrophobic Amino Acids	Basic Amino Acids	Neutral Amino Acids	Acidic Amino Acids	Aromatic Amino Acids
S4	1, 2, 3, 4, 11, 12	6, 7, 13, 15	/	5, 8, 9, 14	10
S2	1, 2, 3, 4, 8, 9, 13, 15	5, 6, 7, 10, 12, 14	/	/	11
S1	2, 13, 15	6, 8, 9, 10, 11, 12	3	7	1, 4, 5, 14

Then, we examine the selectivity of each mutant for the substrates. The S1 and S2 pockets of mutant A are mutated to negatively charged amino acids, while the others remain unchanged. Therefore, 0 and -1 in the substrate should select basic amino acid residues.

CHECKPOINT

1 PTS

The S1 and S2 pockets of mutant A are mutated to negatively charged amino acids

CHECKPOINT

1 PTS

Mutant A prefers positively charged amino acids at positions 0 and -1, and the substrates that meet the conditions are 6, 10, and 12

Referring to the table, substrates 6, 10, and 12 meet the conditions. The other two positions of the substrate should be the same as the optimal substrate of WT. Therefore, mutant A should catalyze substrate number 12 the fastest.

CHECKPOINT

1 PTS

The other two positions of the substrate should be the same as the optimal substrate of the wild-type enzyme

CHECKPOINT

1 PTS

Mutant A catalyzes substrate number 12 the fastest

The S4 pocket of mutant B is mutated to a negatively charged residue, and the S1 pocket is mutated to a hydrophobic Leu with a certain volume of side chain. Therefore, the preferred substrate is -3 positively charged, and 0 is a hydrophobic amino acid.

CHECKPOINT

1 PTS

The S4 pocket of mutant B is mutated to a negatively charged residue, and the S1 pocket is mutated to a hydrophobic residue

Substrates 13 and 15 meet the conditions.

CHECKPOINT

1 PTS

Mutant B prefers substrates with hydrophobic amino acids at position 0 and positively charged amino acids at position -3, and substrates 13 and 15 meet the conditions

The only difference between 13 and 15 is the residue at position 0, one is Ala and the other is Gly. Considering the wild-type sequence, S1 is Gly, and the corresponding substrate is Phe; and the side chain size of Leu is significantly smaller than Phe. At this time, the side chain size of residue 0 should be slightly larger than Gly. In addition, the methyl group of Ala is more conducive to the formation of hydrophobic interactions between side chains than the hydrogen atom of Gly. Therefore, substrate 13 is the most suitable.

CHECKPOINT

1 PTS

Ala at position 0 of substrate 13 is more favorable for interacting with the S1 pocket of mutant B than Gly at position 0 of substrate 15

CHECKPOINT

1 PTS

Mutant B catalyzes substrate number 13 the fastest

Finally, we examine mutant C. The N62V mutation in S2 is a mutation that is difficult to directly assess interactions. We first examine Y104K, which is a mutation of a basic residue.

CHECKPOINT

1 PTS

The S4 pocket of mutant C is mutated to a basic amino acid

Therefore, it is more suitable for the substrate -3 position to be an acidic amino acid. The substrates that meet this condition are 5, 8, 9, 14.

CHECKPOINT

1 PTS

Mutant C prefers substrates with acidic amino acids at position -3, and the substrates that meet the conditions are 5, 8, 9, 14

Since S1 is not mutated, residue 0 is not affected and should be F. Based on this, substrate 8 can be excluded.

CHECKPOINT

1 PTS

S1 is not mutated, and mutant C prefers substrates with F at position 0

The S2 positions of substrates 5, 9, and 14 are K, V, and H, respectively; the WT match is N-P; since the side chain of V is smaller than N in length, from a spatial perspective, something slightly larger than P may be a better substrate. However, if the side chain volume is too large, it may collide with V and affect binding. Based on this, exclude number 5, because the side chain volume of K is too large.

CHECKPOINT

1 PTS

The side chain volume of substrate 5 is too large, and there may be repulsion with the pocket

Next, we examine the side chains of V and H. From the perspective of interaction, the interaction between the positively charged H with the hydrophobic pocket is less favorable than the hydrophobic side chain V; from a spatial perspective, $H > V > P$, H is slightly larger than V. By comparison, substrate 9 is the most suitable for mutant V.

CHECKPOINT

1 PTS

From the perspective of the hydrophobicity matching of the side chain with the S2 pocket of mutant C and the size of the side chain, V at position -1 is more suitable than H

CHECKPOINT

1 PTS

The most suitable substrate for mutant C is substrate number 9

Therefore, the optimal substrates for the three mutants are numbers 12, 13, and 9, respectively. Option O is selected for this question.

CHECKPOINT

1 PTS

Option O is correct